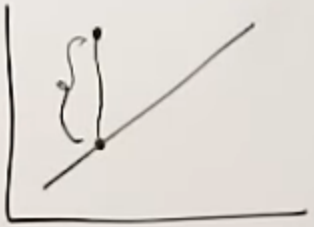
Regularization

→ Ridge

→ Lasso.

(Lasso Regression)



$y = 0.1 + 1.2x_1 + 20x_2 + 39x_3$

$y = 0.9 + 0 \cdot x_1 + 0x_2 + 5x_3$

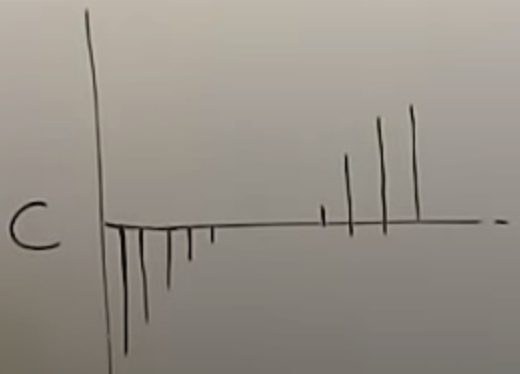
$\Rightarrow$ LassoR = Loss + $\alpha \|W\|$

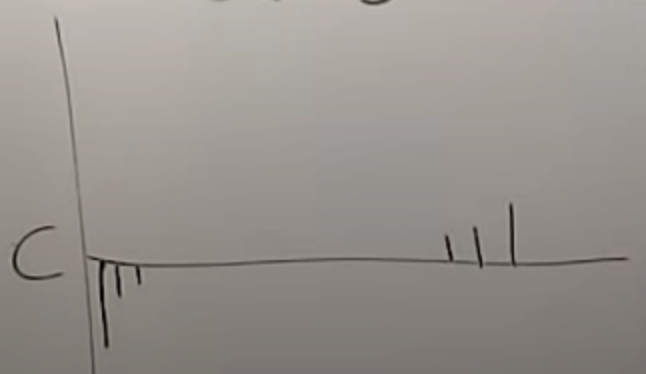
(penalty)

$\|W\| = w_1 + w_2 + w_3 + \cdots + w_n$

$\alpha = 0.001$ $\alpha = 1$ $\alpha = 5$





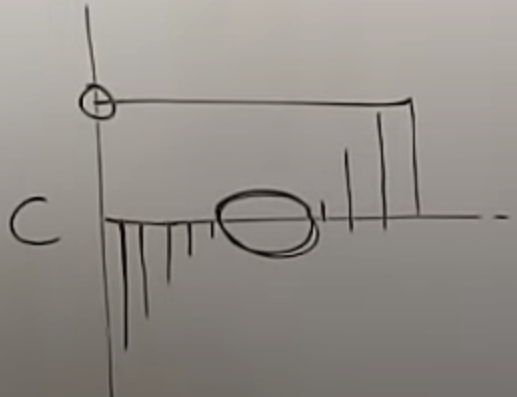


$$LassoR = Loss + \alpha \|W\|$$

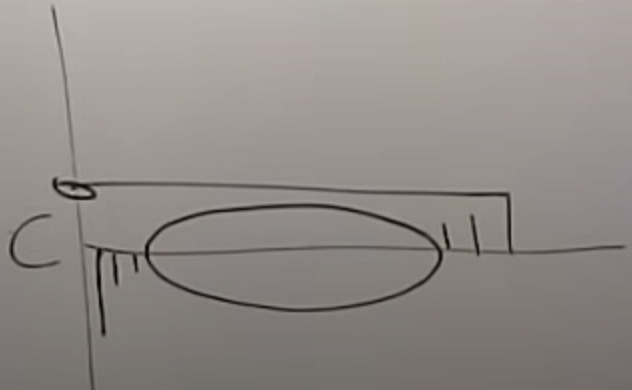
(penalty)

$$\|W\| = |w_1| + |w_2| + |w_3| + \cdots + |w_n|$$

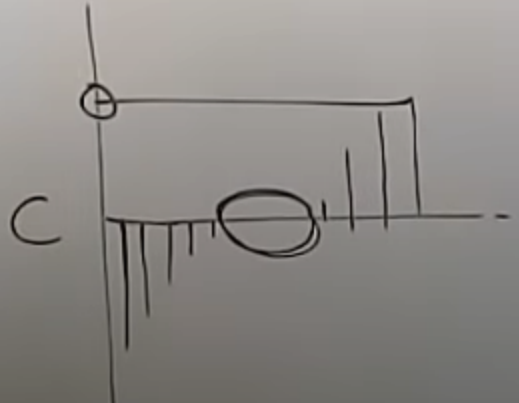
$\alpha = 1$



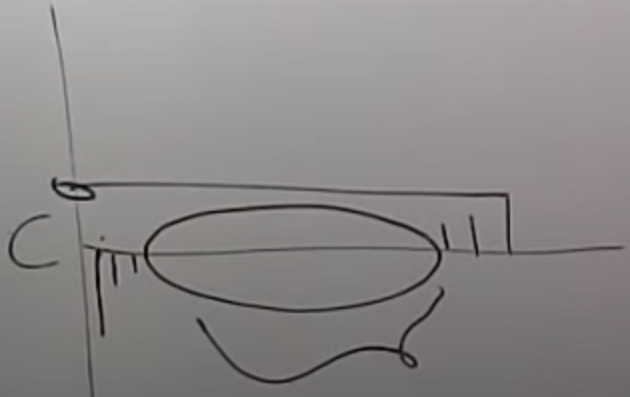
$\alpha = 5$



$\alpha = 1$



$\alpha = 5$



$y = 0.9 + 1.2x_1 + 20x_2 + 39x_3$

$y = 0.9 + 0 \cdot x_1 + 0x_2 + 5x_3$

$y = 3 \cdot 1 + 1.2x_1 + 20x_2 + 39x_3$

$y = 0.9 + 0 \cdot x_1 + 0 \cdot x_2 + 5x_3$

I will $y = 0.9 + 5x_3$

nalty)

$\alpha = 5$

